MIL-S-11846D(ME)
30 October 1984
SUPERSEDING
MIL-S-11846C(ME)
21 March 1975

MILITARY SPECIFICATION

SUPPORT, OVERHEAD-BRACING; FOR PORTABLE PANEL BRIDGE (BAILEY-TYPE)

This specification is approved for use by the Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of overhead-bracing support which acts as a connector for the overhead bracing members of the triple-story, Bailey-type, portable, panel bridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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SPECIFICATIONS

FEDERAL

FF-B-575	- Bolts, Hexagon and Square.
FF-N-836	- Nut; Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat.
FF-W-92	- Washer, Flat (Plain).
QQ-S-781	- Strapping, Steel and Seals.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13220E6847	- Support, Overhead Bracing.

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(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI/ASME	- Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS A 5.1	- Covered Carbon Steel Arc Welding Electrodes.
ANSI/AWS A 5.18	- Carbon Steel Filler Metals for Gas Shielded Arc Welding.
ANSI/AWS D 1.1	- Structual Welding Code.

(Application for copies should be addressed to the American National Standards Institute, Inc. 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The bracing support shall be fabricated as shown on Top Assembly TA13220E6847 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure

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correct fit, assembly, and operation of the bracing support. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and on the drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Material deterioration and control. The bracing support shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.4 Welding electrodes. Welding electrodes shall conform to ANSI/AWS A 5.1 or A 5.18, classification and size as required.

3.5 Bolts and nuts.

3.5.1 Bolts. The bolts shall conform to FF-B-575, type II.

3.5.2 Nuts. The nuts shall conform to FF-N-836, type II, style 4.

3.6 Washers. The washers shall conform to FF-W-92, type A, grade I, class A.

3.7 Identification marking. The bracing support shall be identified in accordance with MIL-STD-130 and as shown on the drawing.

3.8 Treatment and painting. All parts of the bracing support shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, lusterless finish. Unless otherwise specified (see 6.2), color shall match color

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No. 36231 of FED-STD-595. Pins, bolts, nuts, and threaded surfaces shall not be painted but shall be coated with a preservative conforming to MIL-C-16173, grade 2.

3.9 Workmanship.

3.9.1 Steel fabrication. Steel used in the fabrication of the bracing support shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape.

3.9.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed.

3.9.3 Welders and welding.

3.9.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor hall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the perticula ode:

ANSI/ASME, Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS Dl.1, Structural Welding Code, Section 5, Qualification.

Certification shall be available for review by the contracting officer or designated representitive. Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME and ANSI/AWS codes.

3.9.3.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of ANSI/AWS Dl.1 Structural Welding Code. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire-brushed clean.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspections requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article bracing support shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The bracing support shall be examined as specified herein for the following defects:

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Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatment.
105. Dimensions not in accordance with drawings.
106. Parts or components missing.
107. Welding electrodes not as specified.
108. Bolts not as specified.
109. Nuts not as specified.
110. Washers not as specified.
111. Workmanship not as specified.
112. Steel fabrication not as specified.
113. Bolted connections not as specified.
114. Welding certification not available.
115. Defective holes; such as out of position, incorrect size, and burrs not removed.
116. Weld defects such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, slag inclusion, or spatter, or incorrect size.

Minor

201. Treatment and painting not as specified.
202. Identification marking missing or incorrect.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples, selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

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117. Box not as specified for level A or B.
118. Quantities packed together exceed the weight limitation of the box for level A or B.
119. Blocking, bracing and securing not adequate to prevent movement or damage for level A or B.
120. Strapping not as specified for level A or B.
121. Packing not in accordance with the referenced document as specified for commercial.
122. Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. The supports shall be packed together in a close fitting box conforming to PPP-B-621, class 2, style 2; in quantities not to exceed the weight limitation of the box. The supports shall be blocked, braced, and secured within the box in a manner to prevent movement or damage. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be finish B. When specified (see 6.2), strapping shall be finish A.

5.1.2 Level B. Packing shall be as specified in 5.1.1 for level A except boxes shall be class 1 and strapping shall be finish A.

5.1.3 Commercial. The supports shall be packed in accordance with ASTM D 3951.

5.2 Marking (see 6.5).

5.2.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.2.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951; in addition, weight and cube of the pack shall be marked on the shipping container.

6. NOTES

6.1 Intended use. The overhead-bracing support is intended for use in the construction of triple-story, Bailey-type, portable, panel bridges.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and the number of units required (see 3.2).

c. Color required when other than as specified (see 3.8).

d. Degree of packing required (see 5.1).

e. When other than finish B strapping is required (see 5.1).

6.3 First article. When a first article inspection is required, the item will be examined and should be a first produced overhead-bracing support. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.5 Marking. Degrees of preservation have not been included as the supports require no additional protection. For purposes of preservation/packing level marking, the preservation shall be considered level A.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-11846D(ME)	Support, Overhead-Bracing; for Portable Panel Bridge (Bailey-

3a. NAME OF SUBMITTING ORGANIZATION Type)	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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